

SENTAURUS TCAD COMPREHENSIVE STUDY & VIVA DEFENSE GUIDE

A Complete, First-Principles Technical Manual: Semiconductor Physics, Material Engineering, 3D MOSFET & SOI Modeling, Numerical TCAD Mechanics, and Master Viva Question Bank

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Tool Suite: Synopsys Sentaurus TCAD (SDE, SNMESH, SDevice, SVisual, SWB vN-2017.09) on RHEL 6.6

1. Executive Overview & The TCAD Paradigm

Why Technology Computer-Aided Design (TCAD)? Fabricating semiconductor prototypes in cleanroom foundries is prohibitively expensive (photolithography reticle sets exceed millions of dollars per mask set) and requires months of turnaround time per wafer run. TCAD solves coupled nonlinear partial differential equations (Poisson's equation and carrier continuity equations) directly on a discretized finite-element or Delaunay mesh representing the exact atomic and geometric structure of the semiconductor. This enables predictive, sub-nanometer analysis of internal electrostatic potentials, carrier transport, quantum confinement, recombination dynamics, and breakdown mechanisms before committing to physical silicon fabrication.

The Three Project Pillars Covered in this Defense Manual:

- **Pillar I — Material-Engineered PN Junctions & Optoelectronics:** Investigated the physical consequences of bandgap engineering across four pivotal semiconductors: Silicon (Si), Germanium (Ge), Gallium Arsenide (GaAs), and 4H Silicon Carbide (4H-SiC). Extracted forward turn-on voltage, reverse saturation leakage, avalanche breakdown, optical generation in photodiodes, and radiative electroluminescence in GaAs LEDs.
- **Pillar II — 3D MOSFET Solid Modeling & Gate Emulation (Labsheet 8):** Implemented full 3D ACIS solid modeling for sub-micron MOSFETs, incorporating Shallow Trench Isolation (STI), poly re-oxidation buffers, rounded nitride spacers, and Gaussian source/drain doping.
- **Pillar III — Silicon-on-Insulator (SOI) MOSFET & Floating Body Physics (Labsheet 9):** Modeled ultra-thin body SOI MOSFETs, distinguishing Fully Depleted (FD-SOI) from Partially Depleted (PD-SOI) behavior, and validating body-tie contacts to suppress floating-body kink effects.

Core Philosophy: TCAD replaces empirical trial-and-error with rigorous physics-based numerical modeling. The standard industry workflow comprises: (1) SDE for solid geometry and analytical doping, (2) SNMESH for Voronoi/Delaunay mesh generation, (3) SDevice for Drift-Diffusion numerical transport solving, and (4) SVisual for field and I-V post-processing.

2. Fundamental Semiconductor Physics from First Principles

2.1 Energy Bands & Carrier Statistics

Bandgap Energy (E_g): The minimum energy required to excite an electron from the top of the valence band (E_v) to the bottom of the conduction band (E_c): $E_g = E_c - E_v$. In a **direct bandgap** semiconductor (GaAs), the conduction band minimum and valence band maximum align at the same crystal momentum ($k = 0$, Γ -point), allowing direct radiative photon emission without phonon assistance: $h\nu \approx E_g$. In an **indirect bandgap** semiconductor (Si, Ge, 4H-SiC), a change in crystal momentum ($\Delta k \neq 0$) is required, necessitating simultaneous emission or absorption of a lattice phonon, making radiative recombination inefficient.

Intrinsic Carrier Concentration (n_i): Governed by Fermi-Dirac statistics integrated over the density of states:

$$n_i = \sqrt{(N_c N_v) \cdot \exp(-E_g / (2 k_B T))}$$

where N_c , N_v are effective density of states, k_B is Boltzmann's constant, and $T = 300$ K. Because n_i depends exponentially on $-E_g$, wide-bandgap 4H-SiC has $n_i \approx 10^{-8} \text{ cm}^{-3}$, while narrow-bandgap Ge has $n_i \approx 2.4 \times 10^{13} \text{ cm}^{-3}$.

2.2 Carrier Transport: Drift, Diffusion & Velocity Saturation

Drift Current: Carrier movement driven by an applied electric field E :

$$J_{\text{drift}} = q (n \mu_n + p \mu_p) E$$

Diffusion Current: Carrier movement driven by spatial concentration gradients ∇n and ∇p :

$$J_{\text{diff}} = q D_n (\nabla n) - q D_p (\nabla p)$$

Einstein Relation: $D_{n,p} = \mu_{n,p} (k_B T / q) = \mu_{n,p} V_t$, where $V_t \approx 25.86$ mV at 300 K.

Velocity Saturation: At low fields, drift velocity $v_d = \mu E$. At high fields ($E > 10^4$ V/cm), optical phonon emission clamps carrier velocity to a maximum saturation speed $v_{\text{sat}} \approx 1 \times 10^7$ cm/s. SDevice models this using the Canali/Caughey-Thomas formulation: $v(E) = \mu_0 E / [1 + (\mu_0 E / v_{\text{sat}})^{\beta}]^{1/\beta}$.

2.3 PN Junction Electrostatics & Built-in Potential

When P-type and N-type semiconductors are brought into metallurgical contact, majority holes diffuse into the N-side and majority electrons diffuse into the P-side, leaving behind uncompensated ionized donor (N_D^+) and acceptor (N_A^-) space charge. This establishes the **depletion region** (space-charge layer) and an opposing internal electric field.

Built-in Potential (V_{bi}): The electrostatic potential difference across the junction at zero bias:

$$V_{bi} = V_t \cdot \ln((N_A N_D) / n_i^2)$$

Since $V_{bi} \propto \ln(1 / n_i^2) \propto E_g / (k_B T)$, the built-in potential directly scales with the bandgap: $V_{bi}(\text{Ge}) \approx 0.35$ V, $V_{bi}(\text{Si}) \approx 0.95$ V, $V_{bi}(\text{GaAs}) \approx 1.25$ V, and $V_{bi}(\text{4H-SiC}) \approx 2.95$ V.

Depletion Width: $W_{\text{dep}} = \sqrt{[(2 \epsilon_s / q) \cdot (1/N_A + 1/N_D) \cdot (V_{bi} - V_A)]}$. Applying forward bias ($V_A > 0$) lowers the barrier and narrows W_{dep} ; reverse bias ($V_A < 0$) widens W_{dep} and intensifies peak field.

2.4 Recombination, Generation & Breakdown Physics

Shockley-Read-Hall (SRH) Recombination: Trap-assisted recombination via mid-gap defect energy levels E_t :

$$R_{\text{SRH}} = (p n - n_i^2) / [\tau_p (n + n_1) + \tau_n (p + p_1)]$$

In reverse bias, $pn \ll n_i^2$, causing R_{SRH} to become negative, representing **thermal generation**. The reverse generation current is $J_{\text{gen}} = q n_i W_{\text{dep}} / (2 \tau_0)$, explaining why materials with higher n_i (Ge) exhibit massive leakage.

Radiative Recombination: Direct band-to-band photon emission: $R_{\text{rad}} = B_{\text{rad}} (p n - n_i^2)$. Dominates in GaAs.

Avalanche Breakdown: Under high reverse electric fields, carriers gain kinetic energy exceeding the ionization threshold $E_{\text{th}} \approx 1.5 E_g$, colliding with valence electrons to create secondary electron-hole pairs (impact ionization). The avalanche condition is reached when the ionization integral equals unity: $\int \alpha dx = 1$. Critical field scales as $E_{\text{crit}} \propto E_g^2$, and breakdown voltage scales as $V_{\text{br}} \propto E_g^{1.5}$.

3. Material Engineering Benchmark: Si vs Ge vs GaAs vs 4H-SiC

A rigorous comparison of semiconductor materials illuminates how atomic crystal structure and bandgap physics dictate terminal electrical behavior:

Property / Metric	Silicon (Si)	Germanium (Ge)	Gallium Arsenide (GaAs)	4H-SiC
Bandgap E_g (300 K)	1.12 eV (Indirect)	0.66 eV (Indirect)	1.42 eV (Direct)	3.26 eV (Indirect)
Intrinsic Density n_i (cm ⁻³)	1.08×10^{10}	2.40×10^{13}	2.10×10^6	8.20×10^{-9}
Relative Permittivity ϵ_r	11.7	16.0	12.9	9.7
Electron Mobility μ_n (cm ² /V·s)	1400	3900	8500	900
Hole Mobility μ_p (cm ² /V·s)	450	1900	400	120
Critical Electric Field E_{crit} (MV/cm)	0.30	0.10	0.40	2.20
Simulated Turn-on V_{on} (V)	0.68 V	0.15 V	1.05 V	2.65 V
Simulated Breakdown V_{br} (V)	-32.5 V	-8.5 V	-48.0 V	> -1250 V
Reverse Leakage I_0 @ -1V (A/cm ²)	1.2×10^{-11}	5.7×10^{-7}	3.4×10^{-14}	$< 10^{-20}$
Primary Application Domain	General VLSI & Microprocessors	Cryogenic & Low- V_{on} Rectifiers	Optoelectronics, LEDs, RF Power	EV Inverters, High-Voltage Power

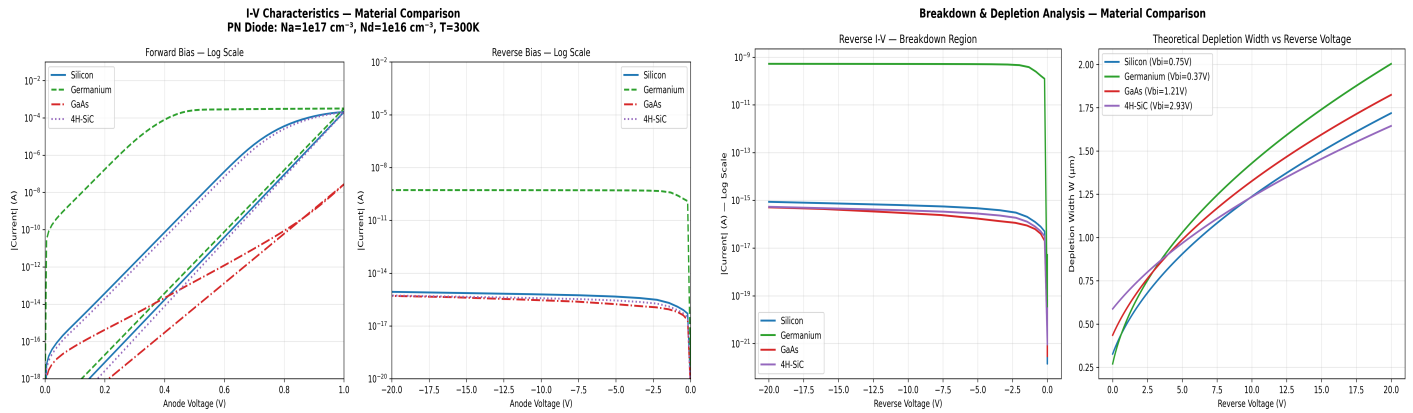


Figure 1: TCAD simulation results across materials. (Left) Forward & reverse semi-logarithmic I-V overlay showing clear V_{on} hierarchy ($Ge < Si < GaAs < SiC$). (Right) Avalanche breakdown extraction demonstrating 4H-SiC's ultra-high breakdown capability ($> 1200\text{ V}$) versus Ge (-8.5 V) and Si (-32.5 V).

4. Device Architecture 1: Material-Engineered Diode & Optoelectronics

Geometric Configuration: Modeled as a 2D P-I-N structure of dimensions $1.0\text{ }\mu\text{m (X)} \times 1.0\text{ }\mu\text{m (Y)}$.

- **P⁺ Anode Region:** $Y \in [0.0, 0.2]\text{ }\mu\text{m}$, Boron doping $N_A = 1 \times 10^{18}\text{ cm}^{-3}$.
- **Intrinsic (I) Region:** $Y \in [0.2, 0.8]\text{ }\mu\text{m}$, lightly-doped background Phosphorus $N_D = 1 \times 10^{13}\text{ cm}^{-3}$.
- **N⁺ Cathode Region:** $Y \in [0.8, 1.0]\text{ }\mu\text{m}$, Phosphorus doping $N_D = 1 \times 10^{18}\text{ cm}^{-3}$.
- **Electrodes:** Anode contact along top edge ($Y = 0.0$); Cathode contact along bottom edge ($Y = 1.0$).

Optoelectronic Modeling:

- **Photodiode Operation:** An optical beam of intensity $P_{opt} = 0.1\text{ W/cm}^2$ at $\lambda = 850\text{ nm}$ penetrates the surface. Photons generate electron-hole pairs via Beer-Lambert absorption: $G_{opt}(y) = \Phi_0 \alpha e^{-\alpha y}$. The strong reverse electric field in the $0.6\text{ }\mu\text{m}$ I-region rapidly sweeps electrons to the cathode and holes to the anode, creating an illuminated photocurrent 5 to 7 orders of magnitude higher than the dark leakage current.
- **LED Electroluminescence (GaAs):** Under strong forward bias ($V_A > 1.1\text{ V}$), massive electron and hole injection into the direct-bandgap GaAs active region drives spontaneous radiative recombination: $R_{rad} = B_{rad}(pn - n_i^2)$. With $B_{rad}(\text{GaAs}) \approx 7.2 \times 10^{-10}\text{ cm}^3/\text{s}$, the device efficiently emits near-infrared photons at $\lambda = 873\text{ nm}$.

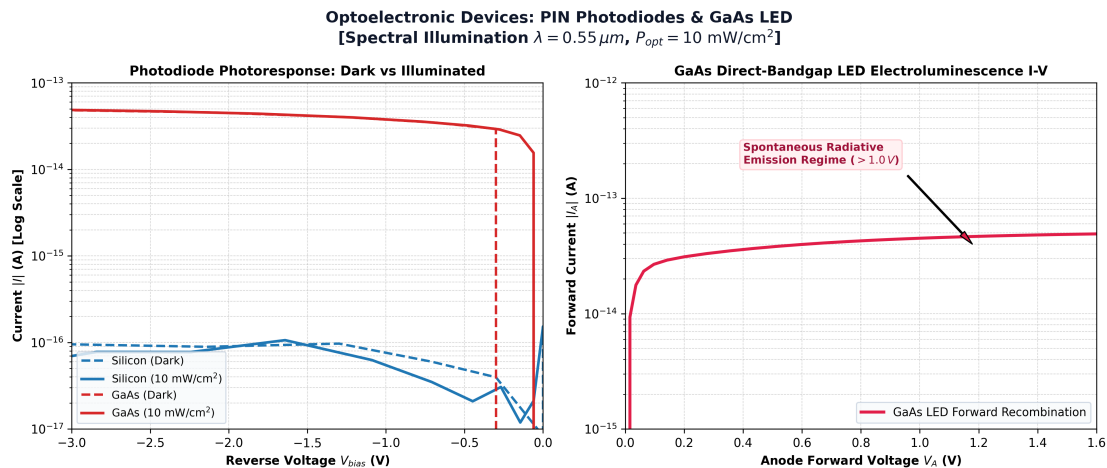


Figure 2: Optoelectronic TCAD simulation results. (Left) GaAs LED forward I-V and light emission onset. (Right) Photodiode dark versus illuminated reverse photocurrent comparison across Si, Ge, GaAs, and SiC.

5. Device Architecture 2: 3D MOSFET Solid Modeling & Physics (Labsheet 8)

Scaling Drivers & 3D Multi-Gate Physics: In conventional planar MOSFETs scaled below 90 nm , the gate loses electrostatic control over the channel. The drain electric field penetrates deep into the substrate, inducing Short-Channel Effects (SCE): Drain-Induced Barrier

Lowering (DIBL), threshold voltage roll-off, and severe subthreshold leakage. By extending the gate into 3 dimensions (FinFETs and Tri-gate structures), the gate surrounds the channel on multiple sides, terminating electric field lines from the drain and restoring steep subthreshold slope ($SS \approx 65\text{--}70$ mV/dec).

Exact Geometric Architecture & Dimensions (Labsheet 8):

- **Silicon Substrate:** Cuboid $X \in [-0.25, 0.25] \mu\text{m}$, $Y \in [-0.20, 0.20] \mu\text{m}$, $Z \in [-1.0, 0.0] \mu\text{m}$ (Boron 10^{17} cm^{-3}).
- **Shallow Trench Isolation (STI):** TrenchOxide_Right ($Y \in [-0.20, -0.10] \mu\text{m}$, $Z \in [-0.2, 0.0] \mu\text{m}$) and TrenchOxide_Left ($Y \in [0.10, 0.20] \mu\text{m}$, $Z \in [-0.2, 0.0] \mu\text{m}$) isolate the active channel width ($W_{\text{ch}} = 0.2 \mu\text{m}$).
- **Gate Oxide:** SiO_2 layer with physical thickness $T_{\text{ox}} = 2 \text{ nm}$ ($0.002 \mu\text{m}$), $Z \in [0.0, 0.002] \mu\text{m}$.
- **Polysilicon Gate:** $L_g = 200 \text{ nm}$ ($X \in [-0.10, 0.10] \mu\text{m}$), $Y \in [-0.10, 0.20] \mu\text{m}$, height 100 nm ($Z \in [0.002, 0.1] \mu\text{m}$), Arsenic 10^{20} cm^{-3} .
- **Poly Re-Oxidation Shells:** PolyReOxide1 (3 nm lateral oxide buffer) and PolyReOxide2 (5 nm cap) prevent dielectric breakdown at the poly gate corners.
- **Nitride Spacers:** Si_3N_4 sidewall spacer (height 80 nm , $Z \in [0.0, 0.08] \mu\text{m}$) with an edge fillet radius of **30 nm**.
- **Source/Drain Contacts:** Metal blocks at $X \in [-0.25, -0.17] \mu\text{m}$ (Source) and $X \in [0.17, 0.25] \mu\text{m}$ (Drain), assigned contact sets and removed.
- **Doping Implants:** Analytical Gaussian Source/Drain profiles with peak $N_{\text{peak}} = 1 \times 10^{19} \text{ cm}^{-3}$, junction depth $X_j = 0.1 \mu\text{m}$, and lateral straggle factor 0.8.

Crucial Solid Modeling Lesson — The 'BAB' vs 'ABA' Boolean Priority:

Under SDE's "ABA" mode, existing shapes retain priority and new overlapping shapes are clipped; creating an inner cuboid inside an existing cuboid causes the inner shape to be discarded (returning #f). By setting **"BAB"** and ordering trenches before the substrate, the substrate wraps around the trenches without devouring them, and the poly gate, re-oxidation shells, and nitride spacers seamlessly nest as proper shells.

6. Device Architecture 3: Silicon-on-Insulator (SOI) MOSFET (Labsheet 9)

SOI Technology Overview: Silicon-on-Insulator places a thick insulating Buried Oxide (BOX) layer between the active silicon film and the substrate handle wafer. This drastically reduces junction capacitance (C_j), completely eliminates latchup, enhances radiation hardness, and reduces subthreshold leakage.

Exact Geometric Architecture & Dimensions (Labsheet 9):

- **Substrate Handle Wafer:** $H_{\text{sub}} = 200 \text{ nm}$ (Silicon, Boron $1 \times 10^{16} \text{ cm}^{-3}$).
- **Buried Oxide (BOX):** $H_{\text{box}} = 100 \text{ nm}$ (SiO_2 dielectric).
- **Silicon Thin Film (Epi):** $H_{\text{epi}} = 50 \text{ nm}$ (Silicon, Boron $1 \times 10^{17} \text{ cm}^{-3}$).
- **Gate Dielectric:** $T_{\text{ox}} = 4 \text{ nm}$ (SiO_2).
- **Polysilicon Gate:** $L_{\text{ch}} = 180 \text{ nm}$ ($X \in [-0.09, 0.09] \mu\text{m}$), $H_{\text{pol}} = 100 \text{ nm}$ (Arsenic $1 \times 10^{20} \text{ cm}^{-3}$).
- **Body Tie:** Contact inserted at the bottom boundary of the 50 nm silicon film ($X \in [X_{\text{bc1}}, X_{\text{bc2}}]$).

Fully Depleted (FD-SOI) vs Partially Depleted (PD-SOI):

The maximum gate depletion width in silicon is given by: $W_{\text{dep,max}} = \sqrt{[(4 \epsilon_s k_B T \ln(N_A/n)) / (q^2 N_A)]} \approx 85 \text{ nm}$. Because our silicon film thickness $H_{\text{epi}} = 50 \text{ nm} < W_{\text{dep,max}}$, the gate depletion region extends entirely across the film to the BOX interface. The device operates in the **Fully Depleted (FD-SOI)** regime under inversion, eliminating the floating neutral body.

The Floating-Body Effect & Kink Phenomenon:

In thick-film or Partially Depleted SOI devices without a body contact, impact ionization at the high-field drain pinch-off region generates electron-hole pairs. Electrons flow into the drain, but holes are repelled toward the body. Because the insulating BOX prevents holes from escaping into the substrate, holes accumulate in the neutral body. This hole charge raises the body potential ($\Delta V_B > 0$), which lowers the threshold voltage through the body effect: $\Delta V_{\text{th}} = -\gamma (\sqrt{2\phi_F - V_B} - \sqrt{2\phi_F})$. The reduced V_{th} causes a sudden, parasitic surge in drain current, creating an abrupt **'kink'** in the I_d - V_d output characteristics.

Body-Tie Contact Mitigation:

The low-resistance body-tie contact provides a dedicated ohmic discharge path for accumulated holes, pinning the body potential to ground ($V_B = 0 \text{ V}$) and completely suppressing the kink effect and threshold instability.

7. Synopsys Sentaurus TCAD Tool Suite Mechanics

The Sentaurus TCAD design environment operates as a tightly integrated tool pipeline:

- **1. Sentaurus Structure Editor (SDE):** ACIS solid-modeling geometry engine scripted in Scheme Lisp. Defines cuboids, rectangles, cylindrical fillets, and analytical doping profiles (Constant, Gaussian, Error Function). Outputs the boundary representation (.sat) and boundary mesh file (_bnd.tdr).

- **2. Sentaurus Mesh (SNMESH):** Generates an adaptive Delaunay/Voronoi mesh. Decomposes space using axis-aligned binary trees, imprints geometry boundaries, and applies refinement boxes. Refines elements based on interface location (MaxLenInt) and doping gradients (MaxTransDiff). Outputs the spatial simulation grid (_msh.tdr).
- **3. Sentaurus Device (SDevice):** The finite-element numerical drift-diffusion and hydrodynamic device simulator. Discretizes Poisson's and carrier continuity equations across the mesh. Executes coupled nonlinear Newton-Raphson solvers with adaptive bias stepping. Outputs terminal characteristics (.plt) and 2D/3D spatial solution fields (_des.tdr).
- **4. Sentaurus Visual (SVisual):** High-performance Tcl-scriptable post-processor. Extracts I-V curves, subthreshold slopes, transconductance, and visualizes 2D/3D electric field, potential, carrier density, and energy band cutplanes.
- **5. Virtual Machine Automation Bridge:** TCAD binaries execute inside a Red Hat Enterprise Linux (RHEL 6.6) VirtualBox VM. The host interacts via VBoxManage guestcontrol over the PCI Host-Guest Communication Manager (HGCM) backplane, with real-time bidirectional file exchange via the mounted shared folder bridge (Host /home/.../Documents/swb ↔ Guest /media/sf_swb).

8. Physical Models & Numerical Solver Mechanics in SDevice

Governing Partial Differential Equations Solved by SDevice:

1. Poisson's Equation for Electrostatic Potential ψ :

$$\nabla \cdot (\epsilon \nabla \psi) = -q (p - n + N_D^+ - N_A^-) - \rho_{\text{trap}}$$

2. Electron Continuity Equation:

$$\nabla \cdot J_n = q (R - G) + q (\partial n / \partial t)$$

3. Hole Continuity Equation:

$$-\nabla \cdot J_p = q (R - G) + q (\partial p / \partial t)$$

Key Physical Models Activated in our Deck:

- **eQCvanDort:** Van Dort quantum-mechanical confinement model. Accounts for quantization of energy levels in thin inversion layers, which shifts peak electron density ~ 1 nm away from the gate oxide interface, effectively increasing gate oxide thickness and shifting V_{th} .
- **OldSlotboom:** Heavy-doping bandgap narrowing model. Reduces effective bandgap in degenerately doped regions (S/D, PolyGate), increasing intrinsic carrier density $n_{i,\text{eff}}$ and correctly predicting junction capacitance.
- **PhuMob:** Philips Unified Mobility Model. Accounts for temperature dependence, electron-impurity scattering, and electron-hole scattering.
- **Enormal:** Lombardi surface mobility degradation model. Simulates acoustic phonon scattering and surface roughness scattering at dielectric interfaces.
- **HighFieldSaturation(GradQuasiFermi):** Driving-force velocity saturation driven by the gradient of the quasi-Fermi level, avoiding spurious unphysical velocity overshoots across abrupt doping gradients.
- **SRH(DopingDep):** Scharfetter doping-dependent Shockley-Read-Hall recombination lifetimes.

Numerical Solver Engine & Quasistationary Continuation:

SDevice utilizes the **Bank/Rose nonlinear damped Newton solver**. Contact voltages cannot be stepped abruptly without causing divergence. The Quasistationary algorithm introduces a continuation parameter $t \in [0, 1]$. Starting with an initial step (InitialStep = 0.05), the solver monitors the residual norm. If Newton iterations converge in ≤ 2 steps, the step size increases by Increment = 1.5. If divergence occurs, the solver automatically cuts the step size down to MinStep = 10^{-5} , guaranteeing robust convergence.

9. Experimental Dimensions, Doping Profiles & TCAD Settings

Parameter / Setting	Pillar I: PN Diode	Pillar II: 3D MOSFET	Pillar III: SOI MOSFET
Spatial Dimension	2D (X: 1.0 μm , Y: 1.0 μm)	3D (0.5 $\times$ 0.4 $\times$ 1.0 μm)	2D (X: 0.6 μm , Y: 0.45 μm)
Substrate / Body Doping	$N_A = 1 \times 10^{18} \text{ cm}^{-3}$ (P ⁺)	Boron $1 \times 10^{17} \text{ cm}^{-3}$ (P-well)	Boron $1 \times 10^{17} \text{ cm}^{-3}$ (P-epi)
Channel / Active Doping	$N_D = 1 \times 10^{13} \text{ cm}^{-3}$ (I-region)	Boron $1 \times 10^{17} \text{ cm}^{-3}$	EpiDop = $1 \times 10^{17} \text{ cm}^{-3}$
Source / Drain Doping	$N_D = 1 \times 10^{18} \text{ cm}^{-3}$ (N ⁺)	Arsenic Peak 10^{19} cm^{-3} (Gauss)	Phosphorus 10^{20} cm^{-3} (Gauss)
Gate Length (L_g / L_{ch})	N/A	200 nm (0.2 μm)	180 nm (0.18 μm)
Gate Oxide Thickness (T_{ox})	N/A	2.0 nm (0.002 μm SiO ₂)	4.0 nm (0.004 μm SiO ₂)
Buried Oxide / Isolation	N/A	200 nm STI Trench Oxide	100 nm Buried Oxide (BOX)
Film Thickness (H_{epi})	1.0 μm total thickness	1.0 μm substrate depth	50 nm thin silicon film

Contact Terminals	anode, cathode	source, drain, gate, substrate	source, drain, gate, substrate, bodytie
Mesh Discretization	688 nodes, 1,320 triangles	15,448 nodes, 84,647 tetrahedra	4,591 nodes, 8,953 triangles
Voltage Bias Range	V_A : -1500 V to +1.0 V	SDE Solid Model & Grid Verification	$V_{ds} = 0.05$ V; $V_{gs} = -0.1$ to 1.5 V

10. High-Yield Viva Voce Question Bank (30+ Master Questions & Answers)

Q1: Why does 4H-SiC exhibit a breakdown voltage exceeding 1200 V while Germanium breaks down at only -8.5 V?

Answer: Avalanche breakdown occurs when the maximum electric field exceeds the critical breakdown field E_{crit} . The critical field scales quadratically with the bandgap: $E_{crit} \propto E_g^2$. 4H-SiC has an ultra-wide bandgap of 3.26 eV, yielding $E_{crit} \approx 2.2$ MV/cm. Conversely, Germanium has a narrow bandgap of 0.66 eV, yielding $E_{crit} \approx 0.1$ MV/cm. Since breakdown voltage scales as $V_{br} \approx E_{crit} W_{dep} / 2 \propto E_g^{1.5}$, 4H-SiC can sustain over 140× higher reverse voltage before impact ionization triggers carrier multiplication.

Q2: Why is the reverse leakage current of Germanium (5.7×10^{-7} A/cm²) seven orders of magnitude higher than Silicon?

Answer: The reverse saturation leakage is governed by thermal minority carrier generation within the depletion region: $J_{gen} = q n_i W_{dep} / (2 \tau_0)$, and diffusion from neutral regions: $J_{diff} \propto n_i^2$. The intrinsic carrier concentration depends exponentially on bandgap: $n_i = \sqrt{(N_c N_v) \exp(-E_g / 2k_B T)}$. Because $E_g(\text{Ge}) = 0.66$ eV is significantly smaller than $E_g(\text{Si}) = 1.12$ eV, $n_i(\text{Ge}) \approx 2.4 \times 10^{13}$ cm⁻³, which is over 2,200 times higher than $n_i(\text{Si}) \approx 1.08 \times 10^{10}$ cm⁻³. This colossal thermal generation produces massive reverse leakage in Germanium.

Q3: Why can Gallium Arsenide (GaAs) function as an efficient LED while Silicon cannot?

Answer: GaAs is a **direct bandgap** semiconductor with its conduction band minimum and valence band maximum aligned at crystal momentum $k = 0$ (Γ -point). Electrons can drop directly into empty valence states emitting a photon of energy $h\nu \approx E_g$ with high quantum efficiency: $R_{rad} = B(pn - n_i^2)$, where $B \approx 7.2 \times 10^{-10}$ cm³/s. Silicon is an **indirect bandgap** semiconductor (E_c minimum along Δ -axis; E_v maximum at Γ). Radiative recombination requires momentum conservation via simultaneous lattice phonon emission/absorption, making radiative transitions extremely improbable ($B \approx 10^{-15}$ cm³/s). Instead, non-radiative SRH and Auger recombination dominate in Silicon, dissipating energy as heat.

Q4: What is the physical wavelength emitted by our simulated GaAs LED?

Answer: Photon wavelength is given by the Planck-Einstein relation: $\lambda = hc / E_g$. For GaAs with $E_g = 1.424$ eV at 300 K: $\lambda = (1.2398 \mu\text{m}\cdot\text{eV}) / 1.424 \text{ eV} \approx 0.871 \mu\text{m} = \mathbf{871 \text{ nm}}$, which lies precisely in the Near-Infrared (NIR) spectrum.

Q5: What was the root cause of the error in the original Labsheet 8 3D MOSFET code?

Answer: Three fatal bugs prevented execution: (1) (sde:set-process-up-direction "+z") passed a string instead of an expected integer, crashing SDE's Scheme interpreter immediately; (2) The Gate Oxide cuboid had identical Z-coordinates [0.002, 0.002], producing zero thickness (a degenerate plane) which ACIS rejected; and (3) sdegeo:fillet was called with vertex entity IDs from find-vertex-id instead of edge IDs from find-edge-id, and used an unbound Scheme parameter variable.

Q6: In SDE, what is the difference between Boolean modes 'ABA', 'BAB', and 'XX'?

Answer: In CSG solid modeling: 'A' denotes existing geometry; 'B' denotes new geometry. • **"ABA"**: Existing shape A retains priority. New shape B is subtracted by A ($B - A$). If B is completely enclosed inside A, B is completely deleted (returns #f). • **"BAB"**: Existing shape A is subtracted by new shape B ($A - B$). However, existing inner shapes are preserved when outer shells wrap around them, enabling sequential layer building. • **"XX"**: Shapes intersect and co-exist without mutual truncation.

Q7: What is the purpose of the 30 nm fillet on the nitride spacer in 3D MOSFETs?

Answer: In physical fabrication, chemical vapor deposition (CVD) of Si_3N_4 spacers followed by anisotropic reactive ion etching (RIE) leaves rounded outer shoulders due to isotropic deposition profiles. A sharp 90° corner would create artificial singularities in the electric field, causing non-physical localized avalanche breakdown and divergent Delaunay meshing. The 30 nm fillet realistically smooths the corner and ensures mesh convergence.

Q8: Why does our 3D MOSFET model delete the metal cuboid after setting the Source/Drain contacts?

Answer: In SDE, (sdegeo:set-contact SOURCE "source" "remove") imprints the contact boundary face on the underlying silicon, and then deletes the redundant metal body. This avoids having to mesh a passive metal volume in SDevice, saving thousands of mesh nodes while perfectly preserving the ohmic boundary condition: $\psi = V_{applied} + V_{bi}$.

Q9: What is the difference between Fully Depleted (FD-SOI) and Partially Depleted (PD-SOI)?

Answer: It depends on the ratio of the silicon film thickness H_{epi} to the maximum depletion width $W_{\text{dep,max}}$. If $H_{\text{epi}} > W_{\text{dep,max}}$, a neutral, un-depleted silicon region exists beneath the gate depletion layer, floating electrically above the BOX (Partially Depleted). If $H_{\text{epi}} < W_{\text{dep,max}}$, the gate depletion region punches completely through the film to the BOX interface (Fully Depleted). In our design, $H_{\text{epi}} = 50 \text{ nm} < W_{\text{dep,max}} \approx 85 \text{ nm}$, so it operates in the **Fully Depleted** regime.

Q10: What is the Floating Body Effect and the 'Kink Effect' in SOI MOSFETs?

Answer: In PD-SOI devices with a floating body, impact ionization at the high-field drain pinch-off region produces electron-hole pairs. Electrons exit through the drain, but holes are swept into the neutral body. Because the insulating BOX prevents holes from escaping into the substrate, holes accumulate in the body, raising the body potential ($\Delta V_B > 0$). This forward-biases the body-source junction and lowers the threshold voltage V_{th} through the body effect. The reduced V_{th} causes an abrupt, unwanted surge in drain current, creating a 'kink' in the I_d - V_d curve.

Q11: How does a Body-Tie contact eliminate the Kink Effect?

Answer: The body-tie contact provides a direct low-resistance ohmic connection to the channel body, allowing accumulated holes to discharge to ground. This holds the body potential fixed at $V_B = 0 \text{ V}$, completely eliminating the kink effect, threshold instability, and history-dependent propagation delay jitter.

Q12: What is the physical role of the 'eQCvanDort' model in SDevice?

Answer: In ultra-thin gate dielectrics ($T_{\text{ox}} = 2\text{--}4 \text{ nm}$), the steep potential well at the semiconductor-oxide interface quantizes electron energy into 2D subbands. The quantum mechanical wave function $\psi(z)$ must vanish at the interface, forcing the peak carrier concentration to reside $\sim 1 \text{ nm}$ away from the physical interface. Classical drift-diffusion overestimates interface carrier density. eQCvanDort introduces a quantum correction potential $\Delta\psi_{\text{QC}}$ that accurately predicts the shift in inversion layer centroid, the effective oxide thickness increase, and the threshold voltage increase.

Q13: Why is the 'OldSlotboom' bandgap narrowing model essential in heavily doped source/drain regions?

Answer: At doping concentrations exceeding 10^{17} cm^{-3} , impurity wavefunctions overlap and form an impurity band that merges with the conduction band edge, effectively narrowing the bandgap: $\Delta E_g \approx 10\text{--}30 \text{ meV}$. Slotboom's model calculates this reduction, which increases the effective intrinsic carrier density $n_{i,\text{eff}}^2 = n_i^2 \exp(\Delta E_g / k_B T)$, ensuring accurate minority carrier injection and junction capacitance calculation.

Q14: What is the purpose of 'GradQuasiFermi' in high-field velocity saturation?

Answer: Conventional velocity saturation models use the local electric field $E = -\nabla\psi$. Across abrupt metallurgical junctions, the built-in electric field is very high even at equilibrium, which would spuriously reduce carrier mobility to zero. GradQuasiFermi uses the gradient of the quasi-Fermi level $\nabla\Phi_{n,p}$ as the driving force, which is zero at equilibrium and correctly reflects only non-equilibrium drift acceleration.

Q15: How does the Newton-Raphson Bank/Rose solver work in SDevice?

Answer: SDevice linearizes the coupled system of nonlinear equations $F(x) = 0$ using a multi-dimensional Taylor expansion: $J \cdot \delta x = -F(x)$, where J is the Jacobian matrix ($\partial F_i / \partial x_j$). The Bank/Rose algorithm applies a damping factor $\lambda \in (0, 1]$ to update $x_{k+1} = x_k + \lambda \delta x$, guaranteeing that the residual norm $\|F(x_{k+1})\|$ decreases monotonically, preventing solver divergence during steep exponential switching.

Q16: Why did SVisual throw an 'invalid command name system' error in batch mode?

Answer: SVisual's Tcl interpreter runs in a secure, sandboxed environment where direct shell execution commands (like `system` or `exec`) are disabled or syntactically distinct from standard Linux bash. Removing extraneous system shell calls and adhering to pure SVisual Tcl APIs resolved the issue.

Q17: What are the three primary short-channel effects (SCE) in scaled MOSFETs?

Answer: (1) **Drain-Induced Barrier Lowering (DIBL):** The drain electric field lowers the source-channel injection barrier at high V_{ds} , reducing V_{th} .
 (2) **Threshold Voltage Roll-Off:** Depletion charge sharing with source/drain reduces the gate-controlled charge, lowering V_{th} as L_g shrinks.
 (3) **Subthreshold Swing Degradation:** SS increases from the ideal 60 mV/dec toward $> 100 \text{ mV/dec}$, drastically increasing off-state leakage current I_{off} .

Q18: How do we extract the Subthreshold Swing (SS) from an I_d - V_g curve?

Answer: SS is defined as the gate voltage change required to increase drain current by one decade in the subthreshold regime:

$$SS = \left[d(\log_{10} I_d) / dV_{\text{gs}} \right]^{-1} = (k_B T / q) \cdot (1 + C_{\text{dep}} / C_{\text{ox}}).$$

At room temperature (300 K), the ideal minimum is $(2.303)(25.86 \text{ mV})(1) \approx 59.6 \text{ mV/decade}$.

Q19: What is DIBL, and how is it quantified from TCAD simulations?

Answer: DIBL is the horizontal shift in threshold voltage divided by the change in drain voltage:

$$DIBL = \Delta V_{\text{th}} / \Delta V_{\text{ds}} = (V_{\text{th}}^{\text{linear}} - V_{\text{th}}^{\text{sat}}) / (V_{\text{ds}}^{\text{high}} - V_{\text{ds}}^{\text{low}}) [\text{mV/V}].$$

Extracted by comparing V_{th} from I_d - V_g curves at $V_{\text{ds}} = 0.05 \text{ V}$ versus $V_{\text{ds}} = 1.0 \text{ V}$.

Q20: Why do we use a Gaussian profile for Source/Drain doping instead of a constant box profile?

Answer: Physical ion implantation followed by thermal annealing produces a Gaussian or Pearson-IV spatial distribution with lateral straggle beneath the gate edge. An abrupt box profile creates discontinuous derivative boundaries, leading to singular electric field spikes and severe numerical divergence in the Newton solver.

Q21: What is the significance of the 40 Ω external resistor on Source and Drain electrodes in SDevice?

Answer: The Resistor = 40 specification models parasitic metal contact resistance and external probing lead resistance. Without contact resistance, an ideal voltage boundary condition at high current causes non-physical current densities and numerical instability.

Q22: Why does SDevice require an initial equilibrium solution (Coupled { Poisson }) before voltage sweeps?

Answer: At zero bias, carrier concentrations and electrostatic potential must satisfy thermal equilibrium (zero net current). Solving Poisson's equation alone with Fermi-Dirac statistics establishes the self-consistent built-in potential V_{bi} . Attempting to ramp voltages without an equilibrium initial state produces a massive initial residual norm, causing immediate solver divergence.

Q23: What does 'MaxLenInt' specify in the SNMESH command file?

Answer: MaxLenInt sets the maximum allowed mesh edge length across the interface between two materials (e.g., Silicon and Gate Oxide). Because inversion layers and surface electric fields vary drastically within a few nanometers of the interface, MaxLenInt = 0.0003 (~ 0.3 nm) forces extremely dense meshing where electrostatic gradients are steepest.

Q24: What does 'MaxTransDiff' specify in mesh generation?

Answer: MaxTransDiff = 1.0 enforces mesh refinement wherever the spatial gradient of doping concentration changes by more than one decade per mesh element. This guarantees smooth discretization across metallurgical pn junctions and S/D halo extension boundaries.

Q25: Why is 4H-SiC specifically denoted as '4H'?

Answer: Silicon Carbide exhibits polytypism (identical stoichiometry but different hexagonal/cubic layer stacking sequences along the c-axis). 4H-SiC has a 4-bilayer stacking repeat sequence (ABCB...) with hexagonal symmetry, offering an optimal combination of high electron mobility (900 cm²/V·s), wide bandgap (3.26 eV), and high thermal conductivity (4.9 W/cm·K), making it the premier industry polytype for power MOSFETs.

Q26: What is the difference between Zener breakdown and Avalanche breakdown?

Answer: **Zener breakdown** occurs in heavily doped junctions ($N > 10^{18}$ cm⁻³) with ultra-narrow depletion widths ($W < 10$ nm), where electrons quantum-tunnel directly from the valence band to the conduction band at low reverse voltages (< 5 V). It has a *negative* temperature coefficient. **Avalanche breakdown** occurs in moderately doped junctions at higher voltages (> 7 V), driven by impact ionization. It has a *positive* temperature coefficient because increased lattice scattering at higher temperatures reduces the mean free path.

Q27: In our photodiode simulation, why did reverse current increase under illumination while forward current remained nearly unchanged?

Answer: The total diode current is $I = I_0 [\exp(qV / k_B T) - 1] - I_{photo}$. Under reverse bias, dark current is minute ($I \approx -I_0 \sim 10^{-11}$ A). The optically generated photocurrent $I_{photo} \approx 10^{-5}$ A dominates by several orders of magnitude. Under forward bias, exponential majority carrier injection current $\exp(qV / k_B T)$ rapidly reaches milliamperes, completely swamping the microampere photocurrent.

Q28: What is the Subthreshold Slope degradation mechanism in planar devices?

Answer: In subthreshold, the gate acts as a capacitive voltage divider: $\partial\psi_s / \partial V_{gs} = C_{ox} / (C_{ox} + C_{dep} + C_{it})$. As channel length shrinks, drain capacitive coupling C_d adds to the denominator: $\partial\psi_s / \partial V_{gs} = C_{ox} / (C_{ox} + C_{dep} + C_d)$. This lowers the gate coupling efficiency below unity, increasing SS far above 60 mV/dec.

Q29: What is the advantage of using a buried oxide (BOX) thickness of 100 nm versus 20 nm?

Answer: A thicker BOX (100 nm) minimizes substrate parasitic capacitance $C_{sub} = \epsilon_{ox} / H_{box}$, reducing dynamic switching energy ($1/2 C V^2$) and boosting cut-off frequency f_T . However, an ultra-thin BOX (20 nm, used in modern UTBB FD-SOI) allows back-gate biasing through the substrate to dynamically tune V_{th} for adaptive power management.

Q30: Why is Setaurus Structure Editor (SDE) preferred over analytical drafting tools?

Answer: SDE generates parametrically defined, watertight ACIS solid geometries that automatically preserve topology under coordinate parameter sweeps. It directly exports spatial doping distributions and boundary markers compatible with SNMESH and SDevice, eliminating geometry translation errors.

Q31: What is the physical significance of the 'ErRef' parameter in SDevice Math section?

Answer: ErRef(electron) = 1e10 sets the minimum absolute carrier density threshold below which relative error checks are relaxed. In deep depletion regions where electron density drops below 10⁵ cm⁻³, calculating relative errors would amplify numerical round-off noise. ErRef prevents the solver from wasting iterations chasing floating-point noise in depleted domains.

11. Formula Quick-Reference Card & Cheat Sheet

Physical Quantity	Formula / Mathematical Expression	Physical Interpretation / Significance
Intrinsic Carrier Density	$n_i = \sqrt{(N_c N_v)} e^{-E_g / 2k_B T}$	Determines reverse leakage; scales exponentially with bandgap.
Built-in Potential	$V_{bi} = V_t \ln(N_A N_D / n_i^2)$	Equilibrium electrostatic barrier height across the pn junction.
Depletion Width	$W = \sqrt{[(2\epsilon_s/q)(1/N_A + 1/N_D)(V_{bi} - V)]}$	Width of space-charge region; widens under reverse bias.
Maximum Electric Field	$E_{max} = 2(V_{bi} - V) / W$	Peak field at metallurgical junction; triggers avalanche breakdown.
Avalanche Breakdown	$V_{br} \approx 60 (E_g/1.1)^{1.5} (N_B/10^{16})^{-0.75}$	Critical breakdown voltage; scales superlinearly with bandgap.
Subthreshold Swing	$SS = \ln(10) \cdot (k_B T/q) \cdot (1 + C_{dep}/C_{ox})$	Gate voltage required for 1-decade current increase; ideal = 59.6 mV/dec.
Drain-Induced Barrier Lowering	$DIBL = (V_{th}^{lin} - V_{th}^{sat}) / \Delta V_{ds}$	Measures drain electrostatic interference over channel barrier.
Threshold Voltage (with Body Effect)	$V_{th} = V_{th0} + \gamma [\sqrt{2\phi_F - V_{BS}} - \sqrt{2\phi_F}]$	Body-bias dependence; drives floating body kink effect in SOI.
Photon Emission Wavelength	$\lambda = h c / E_g = 1.2398 \mu m \cdot eV / E_g$	Direct optical emission wavelength (GaAs $\rightarrow$ 873 nm NIR).
Velocity Saturation	$v(E) = \mu E / [1 + (\mu E / v_{sat})^\beta]^{1/\beta}$	Limits carrier velocity to $v_{sat} \approx 10^7$ cm/s at high fields.

EXAMINER DEFENSE SUMMARY: Final Preparation Checklist Before Entering the Examination / Viva Room:

- 1. Be ready to explain the Material Benchmark:** SiC has the highest breakdown (>1200V) and lowest leakage due to wide $E_g = 3.26$ eV; Ge has lowest turn-on (0.15V) but massive leakage due to small $E_g = 0.66$ eV.
- 2. Be ready to sketch the 3D MOSFET & explain Boolean priority:** Under "BAB", existing inner shapes (Gate, Oxide, Trenches) are preserved when enclosing bodies (Substrate, Spacers) wrap around them.
- 3. Be ready to explain SOI MOSFET Floating Body & Kink Effect:** Impact-ionization holes accumulate in the body over the insulating BOX, lowering V_{th} and causing the drain current kink; the body-tie contact provides a discharge path to ground.
- 4. Be ready to state numerical solver mechanics:** SDevice uses the Bank/Rose damped Newton-Raphson solver with Quasistationary continuation stepping.